

MOHAMMED IRFAN K

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PROFILE

I'm a PhD student in Computer Vision at MBZUAI in Abu Dhabi. My research centers on building multimodal AI systems for **personalized assistance**: systems that can see, listen, remember, reason, and speak.

Interests: omni-modal LLMs, agents and tool use, embedding and retrieval, multimodal reasoning benchmarks, real-time speech-to-speech, multilingual MLLMs, healthcare and assistive AI.

PUBLICATIONS

1. **M. I. Kurpath**, J. M. Kaithakkodan, S. S. Mullappilly, I. Laptev, H. Cholakkal. **Omni-Embed-Mini: Binding Modalities Without Forgetting via Dense Distillation**. **EMNLP 2026 Findings** (to appear). [\[Project Page\]](#)
2. H. Rasheed*, **M. I. Kurpath***, B. Ren, H. Cholakkal, F. S. Khan, S. Khan. **Hard Vision, Easy Vision: What GPT-6 Astra Reveals Across Computer Vision**. **arXiv preprint**, 2026. [\[Project Page\]](#)
3. S. S. Mullappilly*, **M. I. Kurpath***, J. M. Kaithakkodan, O. Mohamed, M. Zidan, F. Khan, S. Khan, R. Anwer, H. Cholakkal. **MediX-R1: Open Ended Medical Reinforcement Learning**. **EMNLP 2026 Findings** (to appear). [\[Project Page\]](#)
4. **M. I. Kurpath***, J. M. Kaithakkodan*, J. Zhou, S. S. Mullappilly, M. Almansoori, N. Ahsan, B. Kalmakhanbet, S. Shikhar, R. Lalla, J. Lahoud, M. Awad, F. S. Khan, S. Khan, R. M. Anwer, H. Cholakkal. **LongShOT: A Benchmark for Omni-Modal Reasoning in Long Videos**. **NeurIPS 2026** (to appear). [\[Project Page\]](#)
5. Y. Kryklyvets, **M. I. Kurpath**, S. S. Mullappilly, J. Zhou, F. S. Khan, R. M. Anwer, S. Khan, H. Cholakkal. **MAviS: A Multimodal Conversational Assistant For Avian Species**. **EMNLP 2025**. [\[Code\]](#)
6. B. S. Shafique, A. Vayani, M. Maaz, H. A. Rasheed, D. Dissanayake, **M. I. Kurpath**, et al., S. Khan, F. S. Khan. **A Culturally-Diverse Multilingual Multimodal Video Benchmark & Model**. **EMNLP 2025**. [\[Project Page\]](#)
7. S. Shikhar, **M. I. Kurpath**, S. S. Mullappilly, J. Lahoud, F. S. Khan, R. M. Anwer, S. Khan, H. Cholakkal. **LLMVoX: Autoregressive Streaming Text-to-Speech Model for Any LLM**. **ACL 2025 Findings**. [\[Project Page\]](#)
8. S. S. Mullappilly*, **M. I. Kurpath***, S. Pieri, S. Y. Alseieri, S. Cholakkal, K. M. Aldahmani, F. S. Khan, R. M. Anwer, S. Khan, T. Baldwin, H. Cholakkal. **BiMediX2: Bio-Medical EXPert LMM for Diverse Medical Modalities**. **EMNLP 2025 Findings**. [\[Code\]](#)
9. **M. I. Kurpath**, P. K. Adwai, J. Bodireddy, C. K. N. S. K. **An IMUs and Potentiometer-Based Controller for Robotic Arm-Hand Teleoperation**. **Sensors and Actuators A: Physical**, 367:115019, 2024. [\[DOI\]](#)

* Equal contribution / joint first authorship.

EXPERIENCE

Aug 2025 to Present	PhD Student, Computer Vision · MBZUAI, Abu Dhabi (advised by Prof. Hisham Cholakkal) Focus: multimodal AI systems for personalized assistance: omni-modal embeddings for memory and retrieval, evidence-seeking agents for long-video understanding, and reliable multimodal models for healthcare. Responsibilities: lead projects end to end, from problem formulation and benchmark design to model training on multi-node GPU clusters, evaluation and paper writing; mentor students and review for CVPR and NeurIPS.
Mar 2024 to Aug 2025	Research Associate · MBZUAI, Abu Dhabi Focus: medical and speech-enabled multimodal LLMs, and multilingual multimodal evaluation. Responsibilities: co-led a bilingual Arabic-English medical LMM, contributed to streaming speech generation for LLMs and to the evaluation toolkit of a multilingual video benchmark; built and presented live demos at GITEX, UNGA79, Abu Dhabi Sustainability Week and Machines Can Think; mentored an MSc thesis and UGRIP interns.
Jul 2022 to Feb 2024	Project Associate · CEET, IIT Madras Research Park, Chennai Focus: multilingual speech and sign-language technology for Indian languages. Responsibilities: built voice-command banking across seven Indian languages (ASR, MT, TTS and transliteration), Indian Sign Language recognition from 3D pose, and a speech-controlled wheelchair guide with a real-time chatbot; mentored summer interns.

AWARDS & GRANTS

2026	Wearable AI Grand Challenge @ ECCV 2026: 1st place , EgoConv track (StreamShot: streaming multi-turn conversation over long egocentric video) and runner-up , EgoLongQA track (Long-ShotV2: long egocentric video QA).
2025	NVIDIA Academic Grant : to advance clinical reasoning capabilities of healthcare LLMs and LMMs. PI: Dr. Hisham Cholakkal.
2025	Meta Regional Research Grant : for Project OMER, building multimodal LLMs for smart wearables. PI: Dr. Hisham Cholakkal.
2024	Meta Llama Impact Innovation Award : for the project BiMediX2, building an Arabic-English bilingual medical LMM. PI: Dr. Hisham Cholakkal.

DEMOS & EXHIBITIONS

- **Personalized wearable assistants for elderly care** with smartwatches, smart glasses, and non-invasive EEG devices, at **GITEX 2025** and **Machines Can Think (MCT) 2026**.
- **Multilingual multimodal LLM integrated with a robotic dog** (Unitree GO2) that holds **interactive conversations** and **carries out requested physical actions** while conversing, at **GITEX 2024** and **Abu Dhabi Sustainability Week 2025**.
- **BiMediX2 assistant** deployed over **Telegram**, presented at the **AI4SD event at UNGA79** and **GITEX 2024**. [\[Demo\]](#)

EDUCATION

2025 to Present	PhD, Computer Vision , MBZUAI	GPA: 3.77/4
2021	B.Tech, Electrical and Electronics Engineering , NIT Puducherry	CGPA: 8.66/10

SERVICE & MENTORING

- **Reviewer**: CVPR, NeurIPS, and Springer Nature journals.
- **Mentor**, **AI for Africa Bootcamp 2026** (MBZUAI × African Leadership Academy, Johannesburg): graduate mentor to high school students.
- **Mentor**, MBZUAI: MSc thesis (MAviS, EMNLP 2025) and UGRIP undergraduate interns.
- **Mentor**, CEET-IIT Madras Research Park: summer interns.

SKILLS

Languages & Tools	Python, Git, Linux
Deep Learning	PyTorch, Hugging Face Transformers, OpenCV, vLLM
Training & Infra	LLM and multimodal LLM training and fine-tuning, distributed and multi-node training, SLURM, AWS HyperPod-class GPU clusters, NVIDIA and AMD clusters
Speech	ASR, TTS, real-time speech-to-speech pipelines
Robotics & Edge	ROS, Unitree GO2 quadruped, NVIDIA Jetson, Raspberry Pi, Arduino, ESP32, embedded sensors, IoT

OTHER PROJECTS

Oct 2024	Real-time Multimodal Conversational Model : speech-in, speech-out over a lightweight VLM, integrated into the Unitree GO2 robot dog and later extended into the personalized wearable assistants for elderly care demonstrated at GITEX 2025 and MCT 2026. [GITEX]
Dec 2022	Fun With Image Processing (Build Lab Course) : an introductory, hands-on OpenCV course for beginners from diverse backgrounds, offered by the Build Club at IITMRP with partner colleges. [GitHub]
Mar 2022	Real-time Expression Recognition : a CNN for seven facial expressions trained on a curated dataset of 9,460 images, reaching 88.35% validation and 82.58% test accuracy, deployed on webcam and video. [GitHub] [Dataset]